

Document 5. Demand, Supply & Market Equilibrium

Topic overview. The price mechanism in a free market allocates scarce resources via three functions — signalling, incentive and rationing — through interaction between demand (DD) and supply (SS). This cheat sheet establishes the canonical framework for explaining any DD/SS question.

Section A. Important Definitions

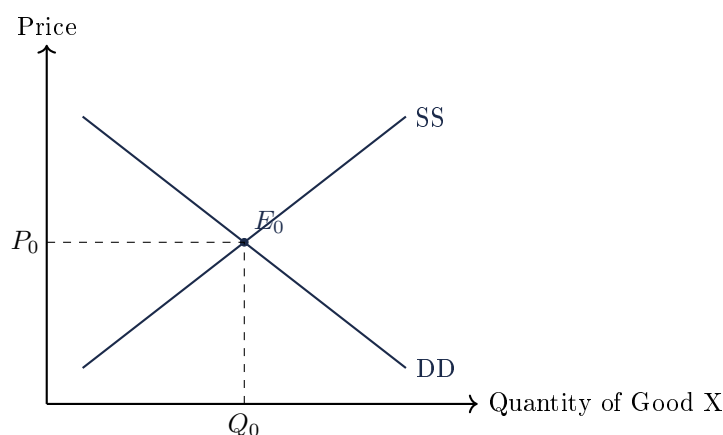
Term	One-line definition
Demand	Quantity of a good consumers are <i>willing and able</i> to buy at each price, <i>ceteris paribus</i> , over a period of time.
Supply	Quantity producers are willing and able to produce at each price, <i>ceteris paribus</i> , over a period of time.
Market equilibrium	The price–quantity combination where $Q_d = Q_s$; no shortage or surplus.
Shortage	$Q_d > Q_s$ at the prevailing price. Creates upward pressure on price.
Surplus	$Q_s > Q_d$ at the prevailing price. Creates downward pressure on price.
Signalling function	Price changes communicate information about scarcity to producers and consumers.
Incentive function	Price changes incentivise producers to enter / exit, and consumers to alter quantity demanded.
Rationing function	Higher prices ration scarce supply to consumers with the highest willingness-to-pay.
Consumer surplus	Area below the demand curve and above the price line.
Producer surplus	Area above the supply curve and below the price line.

Section B. Determinants of Demand and Supply

Determinants of demand (P-TIDE): Price of related goods (substitutes / complements), Taste & preferences, Income (and expectations of income), Demographics (population size and structure), Expectations of future prices.

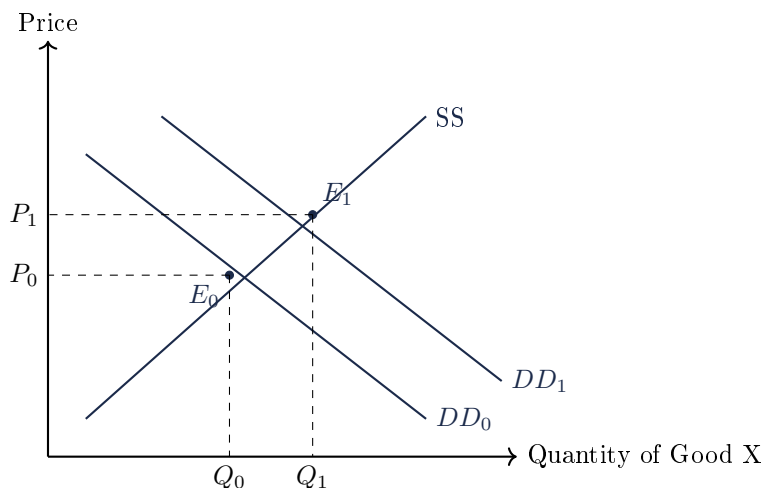
Determinants of supply (CPPSE): Cost of production, Prices of related goods in production, Producers (number of), Supply shocks (weather, war), Expectations of future prices.

Section C. Diagram 1 — Equilibrium



Section D. Question Type 1: Explain the price adjustment process following an increase in demand

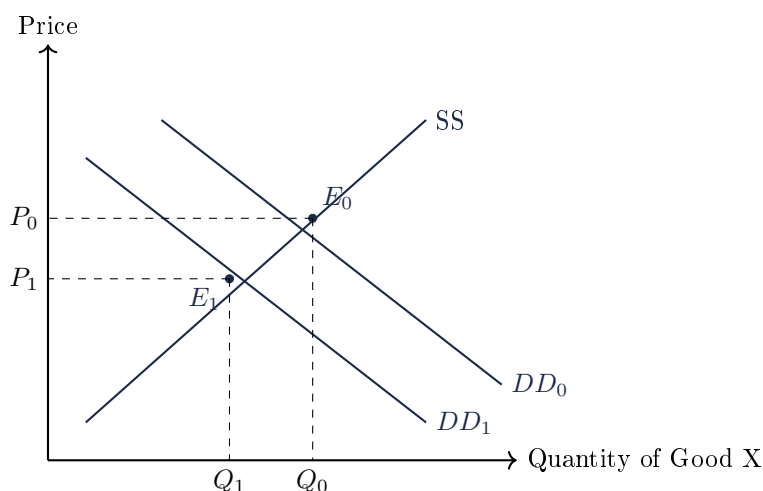
Technique: apply the seven-step Crucible framework. State the trigger, shift, disequilibrium, demand-side adjustment, supply-side adjustment, new equilibrium, and net effect.



- Step 1.** Select the relevant **demand factor (E-G-Y-P-T)** and explain how it affects demand.
Step 2. Consequently, **demand increases**. This is represented by a **rightward shift from DD_0 to DD_1** .
Step 3. Thus, a **shortage of Q_dQ_s** is created. This creates an **upward pressure** on prices.
Step 4. As **prices increase**, quantity demanded decreases along DD_1 to E_1 ; and
Step 5. **quantity supplied increases along SS to E_1** .
Step 6. Via the **price adjustment process**, a new equilibrium is established at E_1 .
Step 7. The **net effect** is an **increase in price by P_0P_1** , and an **increase in quantity by Q_0Q_1** .

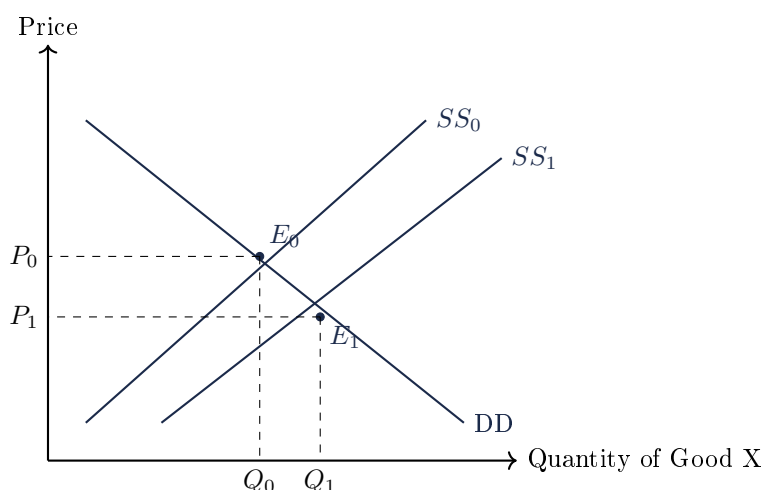
NOTE: Framework reproduced word-for-word from the Crucible Education Centre handout to preserve calibrated language. Substitute the trigger sentence in Step 1 to suit the question context.

Section E. Question Type 2: Decrease in Demand



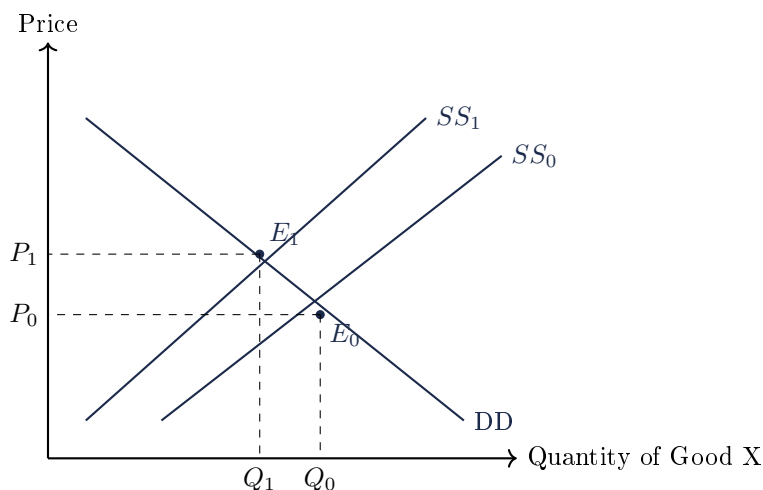
- Step 1.** Select the relevant **demand factor (E-G-Y-P-T)** and explain how it affects demand.
Step 2. Consequently, **demand decreases**. This is represented by a **leftward shift from DD_0 to DD_1** .
Step 3. Thus, a **surplus of Q_dQ_s** is created. This creates a **downward pressure** on prices.
Step 4. As **prices decrease**, quantity demanded increases along DD_1 to E_1 ; and
Step 5. **quantity supplied decreases along SS to E_1** .
Step 6. Via the **price adjustment process**, a new equilibrium is established at E_1 .
Step 7. The **net effect** is a **decrease in price by P_0P_1** , and a **decrease in quantity by Q_0Q_1** .

Section F. Question Type 3: Increase in Supply



- Step 1.** Select the relevant **supply factor (W-E-T-P-I-G-S)** and explain how it affects supply.
Step 2. Consequently, **supply increases**. This is represented by a **rightward shift from SS_0 to SS_1** .
Step 3. Thus, a **surplus of Q_sQ_d** is created. This creates a **downward pressure** on prices.
Step 4. As **prices decrease**, quantity demanded increases along DD to E_1 ; and
Step 5. **quantity supplied decreases along SS_1 to E_1** .
Step 6. Via the **price adjustment process**, a new equilibrium is established at E_1 .
Step 7. The **net effect** is a **decrease in price by P_0P_1** , and an **increase in quantity by Q_0Q_1** .

Section G. Question Type 4: Decrease in Supply



Step 1. Select the relevant **supply factor (W-E-T-P-I-G-S)** and explain how it affects supply.
Step 2. Consequently, **supply decreases**. This is represented by a **leftward shift from SS_0 to SS_1** .
Step 3. Thus, a **shortage of Q_sQ_d** is created. This creates an **upward pressure** on prices.
Step 4. As **prices increase**, quantity demanded decreases along DD to E_1 ; and
Step 5. **quantity supplied increases along SS_1 to E_1** .
Step 6. Via the **price adjustment process**, a new equilibrium is established at E_1 .
Step 7. The **net effect** is an **increase in price by P_0P_1** , and a **decrease in quantity by Q_0Q_1** .

Section H. Question Type 5: Multiple Shifts (Concurrent DD and SS Changes)

When the question tests concurrent changes, the final outcome depends on which factor is more significant.

	DD↑	DD↓	SS↑	SS↓
ΔP	↑	↓	↓	↑
ΔQ	↑	↓	↑	↓

Step 1. Select the relevant **demand factor** and explain how it affects demand. **Demand increases / decreases**. Rightward / leftward shift from DD_0 to DD_1 .
Step 2. Select the relevant **supply factor** and explain how it affects supply. **Supply increases / decreases**. Rightward / leftward shift from SS_0 to SS_1 .
Step 3. To identify the *shortage / surplus*, trace horizontally from P_0 until the dotted line intersects both DD_1 and SS_1 . Q_d is where it intersects DD_1 ; Q_s where it intersects SS_1 .
Step 4. As prices adjust, quantity demanded and quantity supplied adjust along the new curves toward E_1 .
Step 5. Via the **price adjustment process**, a new equilibrium is established at E_1 .
Step 6. Look at the graph to identify the final change in prices and quantity. If you **exaggerated the relative shifts** in each curve, the graphical representation will be accurate.
Step 7. The **net effect** is an increase / decrease in price by P_0P_1 , and an increase / decrease in quantity by Q_0Q_1 .

NOTE: For 4-mark+ parts, candidates must make an evaluative judgement on which factor is the more significant, using the source or contextual knowledge.

Section I. Worked Examples

Worked Example 1: Rise in Income (Normal Good)

Q. Using a diagram, explain how a rise in real disposable income in Singapore affects the market for restaurant meals.

Step 1. Restaurant meals are a normal good. A rise in income increases consumers' ability to purchase normal goods.
Step 2. Demand increases, shown by a rightward shift from DD_0 to DD_1 .
Step 3. A shortage of Q_dQ_s at P_0 creates upward pressure on prices.
Steps 4–5. As prices rise, Q_d falls along DD_1 ; Q_s rises along SS.
Step 6. New equilibrium at E_1 .
Step 7. Net effect: price rises by P_0P_1 ; quantity rises by Q_0Q_1 .

Worked Example 2: Increase in COE Premiums (Cost Shock)

Q. Explain how an increase in COE premiums affects the market for new cars in Singapore.

Step 1. COE is an unavoidable cost of placing a car on the road, so it increases cost of production for car retailers.

Step 2. Supply decreases, shown by a leftward shift from SS_0 to SS_1 .

Step 3. Shortage of $Q_s Q_d$ at P_0 exerts upward pressure on prices.

Steps 4–5. Q_d falls along DD; Q_s rises along SS_1 as prices rise.

Step 6. New equilibrium at E_1 .

Step 7. Net effect: price rises by $P_0 P_1$; quantity falls by $Q_0 Q_1$.

Worked Example 3: Concurrent Shock (Singapore Hawker Food, 2022)

Q. In 2022, food cost inflation raised hawker stall input costs, while consumer wages also rose. Explain the net effect on the market for hawker meals.

Step 1. Demand: higher wages increase income, raising demand for normal goods. $DD_0 \rightarrow DD_1$.

Step 2. Supply: higher food input costs raise the cost of production. $SS_0 \rightarrow SS_1$ (leftward).

Step 3. Both shifts push price up. Whether quantity rises or falls depends on which dominates.

Evaluation. In Singapore, supply-side cost shocks dominated as income gains were modest in real terms (real wage growth $\sim 0\%$ in 2022 due to inflation). Net effect: price rose; quantity fell modestly.